

# Clicks are in the Air: How to Support the Interaction with Floating Objects through Ultrasonic Feedback

Sonja Rümelin, Thomas Gabler, Jesper Bellenbaum

BMW Group

Munich, Germany

{sonja.ruemelin, thomas.gabler, jesper.bellenbaum}@bmw.de

## ABSTRACT

Gestural interfaces are increasingly integrated in human-machine interfaces, but they often rely on visual and auditory feedback and neglect the haptic channel. With technologies providing tactile feedback in midair becoming more mature, it is important to investigate their specific properties. In this paper, we focus on pointing gestures and the combination with floating displays, more precisely tapping buttons. We report the results of a user study investigating in detail different parameters of feedback created through ultrasonic transducers. The results show that the feedback is perceived as most suitable when a modulation frequency of about 150-200Hz is used, which confirms the findings on sensitivity for vibro-tactile stimuli on surfaces. The duration of the signal of either 50 or 130ms did not have a significant effect. Moreover, we present a questionnaire consisting of word pairs to rate the subjective perception of haptic characteristics, to provide a basis for future research.

## Author Keywords

Tactile feedback; ultrasonic transducers; floating buttons.

## CCS Concepts

• Human-centered computing~Haptic devices • Human-centered computing~Gestural input

## INTRODUCTION

With the announcement of the Kinect in 2009, gestural interaction has become a common feature for gaming, TV and mobile applications. Gestures are often inspired by the interaction between humans and make simple tasks such as swiping pages in a presentation easy to use. However, they still haven't taken over control in all domains, which might be due to technical issues or human accuracy. This results in a modality that is intuitive, but lacks the power required for more complex applications [7].

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [Permissions@acm.org](mailto:Permissions@acm.org).

AutomotiveUI '17, September 24–27, 2017, Oldenburg, Germany  
© 2017 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-5150-8/17/09...\$15.00

<https://doi.org/10.1145/3122986.3123010>

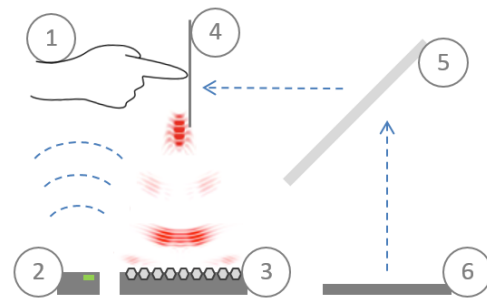


Figure 1. Side view of setup to test ultrasonic feedback for virtual buttons. 1) Interacting hand 2) Gesture sensor 3) Transducer array 4) Virtual image 5) Optical array 6) LCD

But gestural interaction goes beyond swiping. Pointing in midair is a further area for gestural interaction. Using virtual or augmented reality (VR/AR), or new technologies which allow for floating displays, gestural interaction can be combined with virtual objects in free space. This enables a more accurate, and in turn, more powerful way of interaction.

Therefore, the scenario we are looking at in this paper is the interaction with visible, virtual objects, for example, tapping a floating button. We focus on the subjective evaluation of midair feedback and present the results of a user study investigating different parameters of ultrasonic feedback. We also present a questionnaire to describe the ideal characteristics of midair tactile feedback.

## RELATED WORK

### Pointing and virtual objects

Pointing as a specific form of gestural interaction has been widely explored for interaction with remote objects, starting with Bolt's "Put that there" in 1980 [4], but also in the automotive domain [20, 6]. Systems include pointing either with just the hand or a single finger (e.g. [20], Leap Motion<sup>1</sup>), or in combination with additional objects (e.g. Nintendo Wii<sup>2</sup>, Myo<sup>3</sup>, [5]). Those are often incorporated for tracking, but can also support interaction by providing additional input options [1] and tactile feedback [20].

Interacting with virtual objects has been possible for a while in VR environments. In these environments,

<sup>1</sup> <https://www.leapmotion.com/>

<sup>2</sup> <http://www.nintendo.com/wiiu/>

<sup>3</sup> <https://www.myo.com/>

additional force feedback tools such as handheld controllers [2] can be used to make touching objects feel realistic, although the tools themselves are not visible and do not disturb the rendered VR scene. Due to the improvements in AR systems and also the possibility to display free-floating objects without a head-mounted device [16, 8], new approaches for the feedback channel are required.

#### Midair tactile feedback

Therefore, new technologies are explored to provide midair feedback. Aereal creates tactile experiences based on air vortex generation [22], however, with a low accuracy of 8.5cm at 1 meter. Other approaches such as time-varying magnet fields [14], electric arcs [24] or laser plasma [15] have only recently been published as early prototypes. Feedback based on ultrasonic transducers has been introduced in 2009 [13]. Its use for tactile feedback has already been applied in combination with floating visuals [11, 17, 22]. Most recently, it has been used in an automotive context, as an element for in-car interaction [3], highlighting the advantages of an on-demand display with features of well-known touch screens such as direct input.

#### Ultrasonic feedback

Midair feedback using ultrasonic transducers is based on the acoustic radiation pressure of ultrasonic waves [10]. By synchronizing the phase delays of multiple transducers, focus points can be generated which become perceivable by the cutaneous senses of the human skin.

Hoshi et al. [10] have conducted experiments to objectively assess the forces which can be achieved with the phased array focusing technique. A total force of 16mN, which was about 11% of the theoretical calculated value, was based on a not-modulated signal. In contrast, hardware buttons often have force thresholds of 2-6N, indicating that the low force level has to be carefully applied to result in appropriate feedback, especially for a dynamic car environment.

Concentrating on proving feedback to the palm, Wilson et al. [26] present guidelines providing motions, such as to prefer larger distances and longer stimulus durations. They also report that a static point within a grid of 8.5mm wide cells can be localized with an accuracy of 99%, with no effect of the signal duration of either 100 or 1000ms.

Obrist et al. [19] used 250Hz in comparison to 16Hz when investigating vocabulary to describe tactile experiences, based on the fact that the skin's mechanoreceptors which react to low (RA, rapidly adapting) and high (PC, Pacinian Corpuscle) frequencies have their peak sensitivities between 2-40Hz and 250-300Hz. Stimuli were presented for up to 5 minutes. The vocabulary used to describe the sensation was e.g. "coming&going" vs. "constant", indicating that the 16Hz signal was perceived as multiple single stimuli instead of one continuous stimulus.

Obrist et al. [18] discuss an experiment where tactile ultrasonic stimuli are used to communicate emotions, by varying patterns and locations on the hand. Stimuli in the

frequency range of 256Hz in combination with high intensity in different durations were shown to create high arousal. Whether this was positive or negative seems to be depending on the direction of movements, either from palm to finger or vice versa. Short and quick stimuli (200ms) caused high arousal while longer stimulations seem to lead to lower arousal.

Vo and Brewster [25] used a modulation frequency of 250Hz when comparing different feedback types in a finger positioning task. They report that users were significantly faster in locating the correct position when remote visual feedback was enhanced with ultrasonic feedback and that the size of objects can be reduced to about 1cm<sup>2</sup> when both feedback types are available.

#### Research focus

Overall, there has been a lot of work investigating different parameters and properties of ultrasonic midair feedback. There has been work on motion patterns and perception of continuous stimuli. A lot of the related work has used 250Hz signals as a base line, assuming that vibrations created by modulated ultrasonic waves have the same effect as vibrotactile signals on surfaces (e.g. used by [9, 20]). We are not aware of any studies that have investigated the details of parameters for short signals. Since there has not been any variation on using 250Hz as a baseline, we want to especially focus on the variation of the frequency range.

Even though there have been investigations of emotions created by ultrasonic midair feedback, we are not aware of any research on how to assess subjective feedback when midair feedback is used to confirm gestural interaction. Therefore, we evaluated a vocabulary to describe the relevant components of midair feedback.

#### COMPARING SHORT ULTRASONIC MIDAIR STIMULI

We conducted a study to investigate the ideal ultrasonic midair feedback for a virtual button scenario. The focus was on short stimuli supporting tapping interaction, similar to vibro-tactile feedback on touchscreens. We concentrated on subjective feedback, keeping in mind that technology will further evolve and that there is a need to develop methods to describe midair tactile feedback.

#### Apparatus

Our prototype is built from three main components: A *transducer array* consisting of 285 40kHz ultrasonic transducers, a *mirror plate* to create the visuals for the floating objects (both as e.g. in [17]) and a *gesture controller* (Leap Motion) to assess the pointing finger (see Figure 1). On the floating image, we presented buttons with a fixed size of 15x15mm. When a button was activated by "pressing" it in midair, a tactile stimulus was given to the fingertip. Participants were wearing Bose noise cancelling headphones with sounds of the sea. This was done to avoid any interference of the evaluation of stimuli resulting from their acoustic features, which arise from the process of modulating the 40kHz waves.

The prototype was put into a simple seating box consisting of seat, steering wheel and armrest. It was placed in the position of a touch screen right next to the steering wheel. We did not set up a dual-task scenario since we assume that subjective preference is not dependant on if the task is performed alone or in parallel.

### Experimental design

When creating ultrasonic midair feedback, there are three parameters that can be adjusted: modulation frequency, duration of the signal and signal intensity. Since the overall intensity of ultrasonic midair feedback is low, we decided to always set it to the full intensity level. Therefore, we varied *frequency* and *duration* as independent variables. Pretests showed that the signal becomes very weak for frequencies above 300Hz, so we decided for 7 levels of feedback ranging from 0 to 300Hz. Starting from Wilson's 100ms [26] we experimented with different durations: stimuli of less than 50ms were hard to perceive, while more than 130ms tended to be too long, as the finger is already retracted. Therefore, we chose these *lower and upper boundaries as the two levels for duration*.

We selected a 9-point scale to compare the different configurations, ranging from *inferior* to *high-grade*. We chose 9 levels to allow for a highly differentiated evaluation. To assess characteristics of an ideal midair feedback, we decided on 21 bipolar pairs of adjectives (see Table 1). Those were inspired by former evaluations of button sounds [12] and then extended with further categories. We chose a 7-point scale in order to comply with frequently used subjective questionnaires such as the UEQ (User Experience Questionnaire) or AttrakDiff.

|             |             |           |            |
|-------------|-------------|-----------|------------|
| relieved    | insecure    | subtle    | excessive  |
| exhausting  | effortless  | short     | long       |
| clear       | unclear     | roundness | sharp      |
| artificial  | natural     | coarse    | fine       |
| strong      | weak        | concise   | blurred    |
| vibrating   | damped      | boring    | pleasant   |
| troublesome | comfortable | reliable  | unreliable |
| pleasant    | offensive   | untidy    | tidy       |
| loose       | crisp       | fresh     | common     |
| deep        | thin        | sharp     | dull       |
| desirable   | disturbing  |           |            |

**Table 1. Pairs of adjectives to assess the ideal midair feedback**

### Procedure

Overall, the experiment took about 45 minutes per participant. After demographic data was captured, the participant sat in the driver's seat and completed a questionnaire on the characteristics of ideal midair feedback. Subsequently, all 14 different stimuli were presented twice in randomized order, to familiarize participants with the range of feedback. After that, the participant was presented with all 14 stimuli, and had to

rate each one using the 9-point scale. To experience the stimulus, a free-floating button had to be "pressed" which was achieved by tapping using a pointing gesture. This action was repeated at least four times for each rating. In order to sort out participants who were unable to differentiate between the stimuli, every stimulus was rated twice. We determined four as the maximum allowed difference between two ratings for the same stimulus. The order of stimuli was counterbalanced.

### Participants

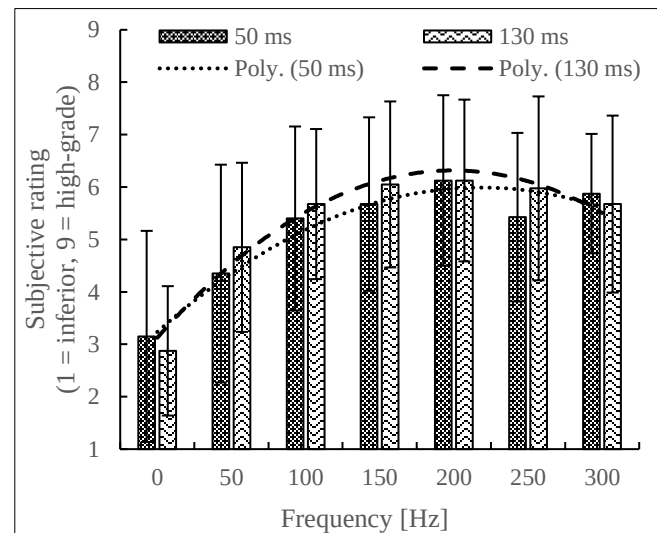
24 participants took part in the study. The average age was 33.9 years, with absolute ages between 20 and 53. 83% were male. One participant was left-handed, but showed no different behavior. 50% of the participants were experts, working in the area of user experience and control elements and the others had no prior experience in this field. Everyone participated voluntarily and did not get paid.

### Results

Four participants (two from each group: experts and novices) had to be excluded from the results because their ratings showed they were not able to perceive differences between the different stimuli. All effects are reported at a .05 level of significance. For the evaluation of the different configurations, a Kolmogorov-Smirnov test showed that the data was normally distributed, so we used an ANOVA with repeated measures to analyze the results.

#### Influence of frequency

Applying a Greenhouse-Geisser correction ( $\epsilon = 0.53$ ) to correct for violation of sphericity, we found a significant influence of frequency on the ratings ( $F(6,19) = 15.3$ , effect size  $f = 0.9$ ). The best ratings were achieved by the stimuli based on 200Hz (6.1 of 9), closely followed by the stimuli based on 150Hz (5.9 of 9). Lower, as well as higher, frequencies received lower ratings (see Figure 2).



**Figure 2. Subjective ratings ("Please rate the perceived tactile quality"), grouped by frequency. Errors bars present the standard deviation. For the purposes of illustration we added a second order polynomial trend line.**

| Pairs of adjectives     | Valence | Precision | Attractiveness | Resolution | Intensity |
|-------------------------|---------|-----------|----------------|------------|-----------|
| exhausting   effortless | -.874   |           |                |            |           |
| sharp   dull            | .846    |           |                |            |           |
| desirable   disturbing  | .828    |           |                |            |           |
| concise   blurred       | .616    |           | .602           |            |           |
| deep   thin             |         | .894      |                |            |           |
| loose   crisp           |         | -.719     |                |            |           |
| clear   unclear         |         | .707      |                |            |           |
| relieved   insecure     |         | .616      |                |            | -.465     |
| pleasant   offensive    |         |           | .888           |            |           |
| boring   pleasant       |         | -.434     | -.605          |            |           |
| artificial   natural    |         |           |                | .865       |           |
| coarse   fine           |         |           |                | .814       |           |
| strong   weak           |         |           |                |            | .892      |

**Table 2. Component matrix, reduced to the most descriptive adjective pairs for midair feedback.**

#### *Influence of duration*

The duration of the stimuli did not have a significant influence on the ratings of the stimuli ( $F(1,19) = 0.43$ ), and there was also no interaction effect between frequency and duration ( $F(6,19)=1.21$ ).

#### *Influence of prior experience*

Unintentionally, half of the participants recruited work in the field of user experience and control elements. Their overall ratings of the stimuli were more differentiated. In particular, their ratings of lower and higher frequencies were more critical, however, this effect was not significant.

#### *Characteristics of midair tactile feedback*

We used a principle component analysis to analyze the ratings of the adjective pairs. Our goal was to reduce the 21 adjective pairs to subordinate properties, explaining the main characteristic features of midair feedback. Eight variables were excluded to achieve a reasonable KMO (Kaiser-Meyer-Olkin) value of 0.61, resulting in five factors describing 78.72% of variances. Table 2 shows the resulting matrix where the remaining 13 most descriptive adjective pairs are grouped to form five main components.

### **DISCUSSION**

Our results show an effect of frequency on the subjective impression of midair tactile feedback created by ultrasonic transducers. Frequencies around 200Hz received the most positive ratings. This indicates that for midair tactile feedback, Pacinian Corpuscles (most sensitive at about 250Hz) are mainly responsible for the sensation of the signals. There is a tendency that the frequency peak is slightly lower than expected, indicating that there might be further aspects that influence the perception. The duration of the signal of either 50 or 130ms did not influence the results, both were perceived as sufficient for a short confirmation.

To characterize midair feedback, we identified the main categories *valence*, *precision*, *attractiveness*, *resolution*,

and *intensity* to fit the matching adjective pairs. Those have to be validated in future studies, and can then be used as relevant properties to define the ideal profile for midair feedback, independent of the technology applied.

With regards to our sample, we did not find differences between experts and novices, indicating that the basic perception of tactile feedback does not change with increasing awareness. However, we found participants who could not discriminate between the different stimuli. Therefore, we want to highlight that it is important to set up evaluations of haptics carefully to allow for detecting random ratings which would distort the results.

### **LIMITATIONS**

As people get older, tactile perceptions deteriorates. We did not design this study to focus on this aspect, but it will be very important to evaluate if this has an effect on the different haptic parameters, and if the feedback will be strong enough to support different age groups and people with physical disabilities.

### **CONCLUSION AND FUTURE WORK**

We have confirmed that midair tactile feedback created by ultrasonic transducers can be oriented towards the objective guidelines of vibro-tactile feedback, when it comes to the perceptibility of frequencies. We presented a set of adjective pairs which can be used for further development of subjective guidelines for midair feedback, and discussed aspects regarding the samples for perception studies.

As a next step towards a good user experience, we want to further develop automotive use cases that benefit from the use of midair tactile feedback, especially when combining gestural interaction with free-floating displays. Furthermore, we plan to investigate its effect compared to other concepts such as auditory feedback or interaction on touch screens in a dual-task scenario, to see whether it helps to reduce distraction while driving.

## REFERENCES

1. Patrick Baudisch, Mike Sinclair, Andrew Wilson. Soap: a pointing device that works in mid-air. In *Proceeding of the 19th annual ACM symposium on User interface software and technology* (UIST '06), 43-46. <https://doi.org/10.1145/1166253.1166261>
2. Hrvoje Benko, Christian Holz, Mike Sinclair, Eyal Ofek. NormalTouch and TextureTouch: High-fidelity 3D Haptic Shape Rendering on Handheld Virtual Reality Controllers. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (UIST '16), 717-728. <https://doi.org/10.1145/2984511.2984526>
3. BMW Group. The BMW Group at CES 2017 in Las Vegas. BMW HoloActive Touch: An innovative operating concept for the interior of the future. Press release 15.12.2016. <https://www.press.bmwgroup.com/global/article/detail/T0266649EN/the-bmw-group-at-ces-2017-in-las-vegas-bmw-holoactive-touch-an-innovative-operating-concept-for-the-interior-of-the-future?language=en>
4. Richard A. Bolt. 'Put that there': Voice and Gesture at the Graphics Interface. In *Proceedings of the 7th annual conference on Computer graphics and interactive techniques* (SIGGRAPH '80), 262-270. <https://doi.org/10.1145/800250.807503>
5. Xiang Cao and Ravin Balakrishnan. VisionWand: interaction techniques for large displays using a passive wand tracked in 3D. In *Proceedings of the 16th annual ACM symposium on User interface software and technology* (UIST '03), 173-182. <https://doi.org/10.1145/964696.964716>
6. Kikuo Fujimura, Lijie Xu, Cuong Tran, Rishabh Bhandari, Victor Ng-Thow-Hing. Driver Queries Using Wheel-Constrained Finger Pointing and 3-D Head-up Display Visual Feedback. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (AutoUI '13), 56-62. <https://doi.org/10.1145/2516540.2516551>
7. Lee Garber. Gestural Technology: Moving Interfaces in a New Direction. *Computer* 46, 10 (2013): 22-25. <https://doi.org/10.1109/MC.2013.352>
8. Daniel Gotsch, Xujing Zhang, Jesse Burstyn, Roel Vertegaal. HoloFlex: A Flexible Holographic Smartphone with Bend Input. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (CHI EA '16), 3675-3678. <https://doi.org/10.1145/2851581.2890258>
9. Eve Hoggan and Stephen Brewster. Designing audio and tactile crossmodal icons for mobile devices. In *Proceedings of the international conference on Multimodal interfaces* (ICMI '07), 162-169. <https://doi.org/10.1145/1322192.1322222>
10. Takayuki Hoshi, Masafumi Takahashi, Takayuki Iwamoto, Hiroyuki Shinoda. Noncontact Tactile Display Based on Radiation Pressure of Airborne Ultrasound. In *IEEE Transactions on Haptics* 3, 3 (2010): 155-165. <https://doi.org/10.1109/TOH.2010.4>
11. Takayuki Hoshi, Masafumi Takahashi, Kei Nakatsuma, Hiroyuki Shinoda. Touchable Holography. In *ACM SIGGRAPH 2009 Emerging Technologies* (23). <https://doi.org/10.1145/1597956.1597979>
12. Shunsuke Ishimitsu, Koji Sakamoto, Takayuki Arai, Toshikazu Yoshimi, Yuichi Fujimoto, Kenichi Kawasaki. A study of evaluating the button sounds. *The Journal of the Acoustical Society of America* 123, 5 (May 2008), 3153-3158. <http://dx.doi.org/10.1121/1.2934134>
13. Takayuki Iwamoto, Mari Tatzono, Hiroyuki Shinoda. Non-contact Method for Producing Tactile Sensation Using Airborne Ultrasound. In *Proceedings of the 6th international conference on Haptics: Perception, Devices and Scenarios* (EuroHaptics '08), 504-513. [https://doi.org/10.1007/978-3-540-69057-3\\_64](https://doi.org/10.1007/978-3-540-69057-3_64)
14. Hyung-Sik Kim, Seong-Young Gim, Woo-Ram Kim, Mi-Hyun Choi, Seungmoon Choi, Soon-Cheol Chung. MagTacS: Delivering Tactile Sensation over an Object. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (UIST '16 Adjunct), 41-42. <https://doi.org/10.1145/2984751.2985698>
15. Hojin Lee, Hojun Cha, Junsuk Park, Seungmoon Choi, Hyung-Sik Kim, Soon-Cheol Chung. LaserStroke: Mid-air Tactile Experiences on Contours Using Indirect Laser Radiation. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (UIST '16 Adjunct), 73-74. <https://doi.org/10.1145/2984751.2985708>
16. Satoshi Maekawa, Kouichi Nitta, Osamu Matoba. Transmissive Optical Imaging Device with Micromirror Array. In *Proceedings Three-Dimensional TV, Video, and Display V* (SPIE 6392), 63920E. <http://dx.doi.org/10.1117/12.690574>
17. Yasuaki Monnai, Keisuke Hasegawa, Masahiro Fujiwara, Kazuma Yoshino, Seki Inoue, Hiroyuki Shinoda. HaptoMime: mid-air haptic interaction with a floating virtual screen. In *Proceedings of the 27th annual ACM symposium on User interface software and technology* (UIST '14), 663-667. <https://doi.org/10.1145/2642918.2647407>
18. Marianna Obrist, Sriram Subramanian, Elia Gatti, Benjamin Long, Thomas Carter. Emotions Mediated Through Mid-Air Haptics. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (CHI '15), 2053-2062. <http://dx.doi.org/10.1145/2702123.2702361>

19. Marianna Obrist, Sue Ann Seah, Sriram Subramanian. Talking about tactile experiences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '13), 1659-1668.  
<http://dl.acm.org/citation.cfm?id=2466220>
20. Sonja Rümelin, Chadly Marouane, Andreas Butz. Free-hand pointing for identification and interaction with distant objects. In *Proceedings of the 5th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (AutoUI '13), 40-47.  
<https://doi.org/10.1145/2516540.2516556>
21. Sonja Rümelin, Hardy, R., and Rukzio, E. NaviRadar: A Tactile Information Display for Pedestrian Navigation. In *Proceedings of the 24th annual ACM symposium on User interface software and technology* (UIST '11), 293-302.  
<https://doi.org/10.1145/2047196.2047234>
22. Antti Sand, Ismo Rakkolainen, Poika Isokoski, Jari Kangas, Roope Raisamo, Karri Palovuori. Head-mounted display with mid-air tactile feedback. In *Proceedings of the 21st ACM Symposium on Virtual Reality Software and Technology* (VRST '15), 51-58.  
<https://doi.org/10.1145/2821592.2821593>
23. Rajinder Sodhi, Matthew Glisson, Ivan Poupyrev. AIREAL: tactile gaming experiences in free air. In *ACM SIGGRAPH 2013 Emerging Technologies* (2).  
<https://doi.org/10.1145/2503368.2503370>
24. Daniel Spelmezan, Deepak Ranjan Sahoo, Sriram Subramanian. Sparkle: Towards Haptic Hover-Feedback with Electric Arcs. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (UIST '16 Adjunct), 55-57.  
<https://doi.org/10.1145/2984751.2985702>
25. Vo Dong-Bach and Stephen Brewster. Touching the invisible: Localizing ultrasonic haptics cues. In *Proceedings of IEEE World Haptics Conference* (WHC '15).  
<https://doi.org/10.1109/WHC.2015.7177740>
26. Graham Wilson, Thomas Carter, Sriram Subramanian, Stephen A. Brewster. Perception of ultrasonic haptic feedback on the hand: localisation and apparent motion. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '14), 1133-1142. <https://doi.org/10.1145/2556288.2557033>